

jQuery: styles, AJAX

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Contents

- Inline CSS classes addition and removal
- Inline CSS modification
- Dimensions
- Animations
- AJAX

jQuery CSS classes

- Get and set CSS classes
- With jQuery it is easy to manipulate the style of elements.
- jQuery Manipulating CSS
 - jQuery has several methods for CSS manipulation:
 - `addClass()`
 - Adds one or more classes to the selected elements
 - `removeClass()`
 - Removes one or more classes from the selected elements
 - `toggleClass()`
 - Toggles between adding/removing classes from the selected elements
 - `css()`
 - Sets or returns the style attribute

Example stylesheet

- Example:

```
.important {  
    font-weight: bold;  
    font-size: xx-large;  
}
```

```
.blue {  
    color: blue;  
}
```

jQuery addClass() Method

- Example:
 - shows how to
 - add class attributes to different elements.
 - you can select multiple elements, when adding classes:
 - `7-jquery-manipulate-css-1.html`

jQuery addClass() Method

- You can also specify
 - multiple classes within the `addClass()` method:
- Example:
 - `8-jquery-manipulate-css-2.html`

jQuery removeClass() Method

- Example shows how to
 - remove a specific class attribute from different elements::
- Example:
 - `9-jquery-manipulate-css-3.html`

jQuery toggleClass() Method

- Example shows how to
 - how to use the jQuery toggleClass() method.
 - This method toggles between adding/removing classes from the selected elements:
- Example:
 - `10-jquery-manipulate-css-4.html`

jQuery css() Method

- jQuery css() Method
 - sets or returns one or more style properties for the selected elements.
- Return a CSS Property
 - To return the value of a specified CSS property:
 - `css("propertyname");`
- Example:
 - return the background-color value of the FIRST matched element:
 - `4-jquery-css-1.html`

jQuery css() Method

- Set a CSS Property
 - To set a specified CSS property:
 - `css("propertyname", "value");`
- Example:
 - set the background-color value for ALL matched elements:
 - `5-jquery-css-2.html`

jQuery css() Method

- Set Multiple CSS Properties

- To set multiple CSS properties:

- `css({ "propertyname": "value", "propertyname": "value", ... });`

- Example:

- set a background-color and a font-size for ALL matched elements::
 - `6-jquery-css-3.html`

jQuery hide() & show() methods

- hide and show HTML elements with the
 - hide() and show() methods:
- Example:

```
$("#hide").click(function() {  
    $("p").hide();  
});
```

```
$("#show").click(function() {  
    $("p").show();  
});
```

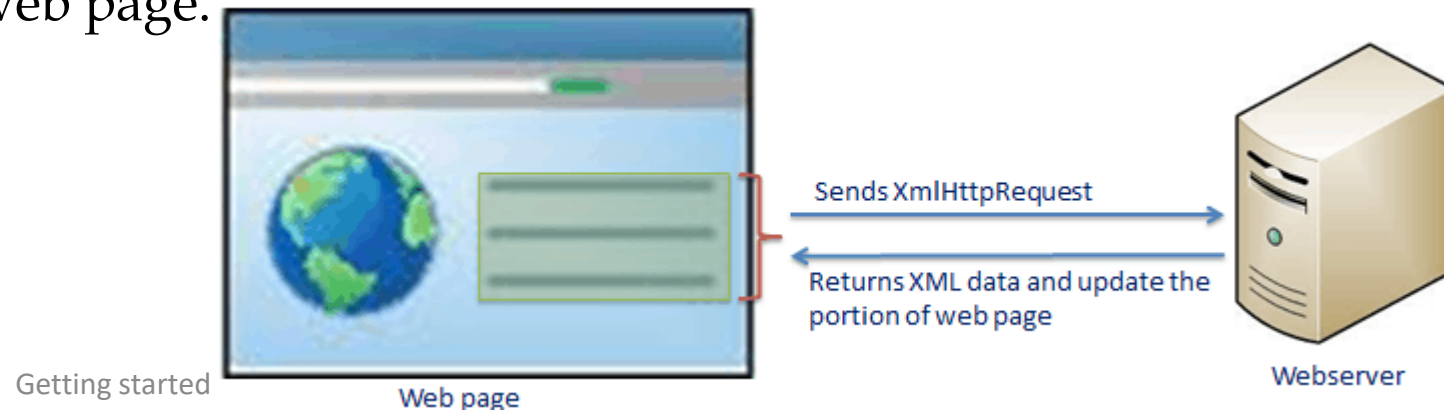
- 1-hide-all-paragraphs-1.html
- 2-hide-all-paragraphs-2.html
- 3-hide-all-paragraphs-3.html

jQuery hide() & show() methods

- hide and show HTML elements with the
 - hide() and show() methods

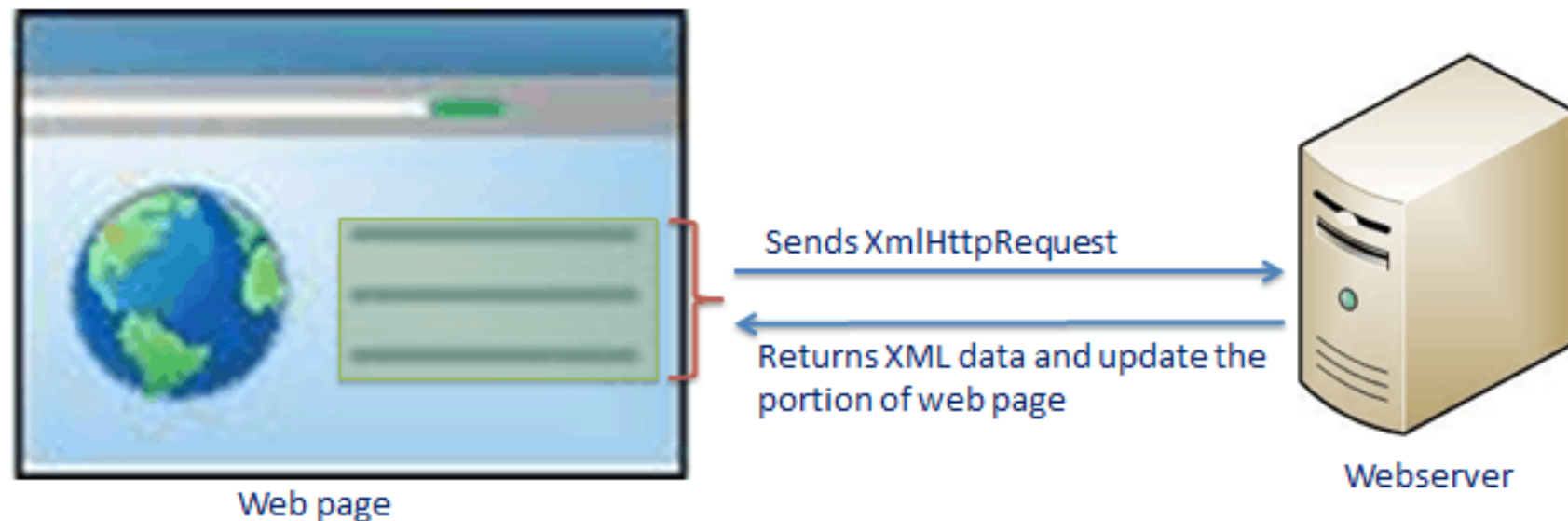
Working AJAX with jQuery

- AJAX
 - Asynchronous JavaScript and XML
- JavaScript includes features of
 - sending asynchronous http request using XMLHttpRequest object.
- AJAX is about using this ability of JavaScript to
 - send asynchronous http request and
 - get the xml data as a response (also in other formats), and
 - update the part of a web page (using JavaScript) **without**
 - reloading or refreshing entire web page.



What is AJAX?

- AJAX
 - Asynchronous JavaScript and XML
 - A mechanism for making partial page updates.
 - It enables us to
 - update sections of page with data that comes from the server;
 - avoid full page refresh.
- YouTube
- Google, GMail
- Google maps
- Google docs, sheets
- FB, Flickr



What is AJAX?

- AJAX
 - **A**ynchronous
 - Multiple AJAX requests work simultaneously
- In synchronous web-requests R1, R2, R3,...
 - R2 wait for completion of R1
 - R3 wait for completion of R2
 - Entire page will be blocked
- In **a**synchronous web-requests R1, R2, R3,...
 - R1, R2 and R3 work simultaneously.
 - No one waits for others' completion.

What is AJAX?

- The **core principle** of AJAX is:
 - **Communicate with the server in**
 - the background without
 - reloading the entire web page.
- Before AJAX:
 - Every server request → **full page reload**
 - Poor user experience (flicker, waiting, lost scroll position)
- With AJAX:
 - Browser sends a request **in the background**
 - Server responds with **data only** (not a whole page)
 - JavaScript updates **only the required part of the page**

What is AJAX?

- **In simple terms:**
 - AJAX allows **partial page updates**, not whole-page refreshes.
- **Key requirements for AJAX:**
 - **JavaScript:** to send requests & update UI
 - **Server endpoint:** to respond with data (JSON/XML/text)
 - **XMLHttpRequest / Fetch API:** communication bridge
 - **DOM manipulation:** update content dynamically
- **Few more Real-world examples:**
 - Live search suggestions
 - Like/Comment counters updating instantly
 - Form validation while typing
 - Gmail loading new emails without refreshing

What is AJAX?

- What does “Asynchronous” *exactly* mean?
 - It is the heart of AJAX.
- **Synchronous (blocking behavior)**
 - Think of it as a **single-lane road**:
 - Request sent to server
 - **Browser waits**
 - Response comes back
 - Page updates
 - User can interact again
 - Problem:
 - User interface is **frozen** until the response arrives.
- **Asynchronous (non-blocking behavior)**
 - Think of it as **multi-lane traffic**:
 - Request sent to server
 - Browser **does NOT wait**
 - User continues scrolling, clicking, typing
 - Server responds **later**
 - Callback / Promise handles the response
 - Page updates **only where needed**
 - UI remains **responsive**

What is AJAX?

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Use AJAX with..

- AJAX
- Not only XML, but also
- JSON
- Normal text data
- php
- Jsp
- ...

Working AJAX with jQuery..

- jQuery library includes various methods to
 - send Ajax requests.
 - These methods internally use XMLHttpRequest object of JavaScript.

jQuery Ajax Methods	Description
ajax()	Sends asynchronous http request to the server.
get()	Sends http GET request to load the data from the server.
post()	Sends http POST request to submit or load the data to the server.
getJSON()	Sends http GET request to load JSON encoded data from the server.
getScript()	Sends http GET request to load the JavaScript file from the server and then executes it.
load()	Sends http request to load the html or text content from the server and add them to DOM element(s).

Working AJAX with jQuery..

- jQuery library also includes events which will be
 - fired based on the state of the Ajax request.

jQuery Ajax Events	Description
<code>ajaxComplete()</code>	Register a handler function to be called when Ajax requests complete.
<code>ajaxError()</code>	Register a handler function to be called when Ajax requests complete with an error.
<code>ajaxSend()</code>	Register a handler function to be called before Ajax request is sent.
<code>ajaxStart()</code>	Register a handler function to be called when the first Ajax request begins.
<code>ajaxStop()</code>	Register a handler function to be called when all the Ajax requests have completed.
<code>ajaxSuccess()</code>	Register a handler function to be called when Ajax request completes successfully.

Working AJAX with jQuery..

- Why jQuery – AJAX?
 - Cross browser support
 - Simple methods to use
 - Ability to send GET and POST requests
 - Ability to load JSON, XML, HTML or scripts

Sending AJAX request

```
var request = new XMLHttpRequest();
```

```
request.open("GET", "URL");
```

```
request.send();
```

jQuery ajax() method

- jQuery ajax() method:
 - provides core functionality of Ajax in jQuery. It sends asynchronous HTTP requests to the server.
 - Syntax:
 - `$.ajax(url);`
 - `$.ajax(url, [options]);`
 - url: A string URL to which you want to submit/retrieve the data
 - options: Configuration options for Ajax request.
 - It is optional and it can be specified using JSON format.
- Examples:
 - 18-ajax-1.html
 - 19-ajax-2.html

jQuery ajaxsetup() method

- jQuery ajaxsetup() method:

```
$.ajaxSetup( {name:value, name:value, ... } )
```

- Examples:
 - 20-ajaxsetup-1.html
 - 21-ajaxsetup-2.html

jQuery getJSON() method

- jQuery getJSON () method:

- fetches JSON-encoded data from the server using a GET HTTP request.

`$(selector).getJSON(url, data, success(data, status, xhr))`

- url: used to specify the URL in the form of a string to which the request is sent.
- data: [optional]. Used to specify data that will be sent to the server.
- callback: [optional]. It runs when the request succeeds.

- Return Value: XMLHttpRequest object.

- Examples:

- 22-ajax-getjson-1.html

jQuery ajaxStart() method

- jQuery ajaxStart () method:

- to specify a function to be run when an AJAX request starts.

`$ (document) .ajaxStart (function () )`

- function: It specifies the function to run when the Ajax request starts.

- Examples:

- 23-ajax-ajaxstart-1.html

jQuery ajaxStop() method

- jQuery ajaxStop() method:
 - to specify a function to be run when an AJAX request has been completed.
`$(document).ajaxStop(function())`
 - function: It specifies the function to run when the Ajax request has been completed.
- Examples:
 - 24-ajax-ajaxstop-1.html

jQuery ajaxSuccess() method

- jQuery ajaxSuccess () method:
 - to specify a function to be run when an AJAX request has been completed successfully.

```
$(document).ajaxSuccess(function(event, xhr, options))
```

- event: event object.
- xhr: XMLHttpRequest object.
- options: the used options in AJAX request.

- Examples:
 - 25-ajax-ajaxSuccess-1.html